

Ilera-ATLAS: technical documentation

NAIC 2026 · Academia & Research Track · Problem Statement 3 · Repository: github.com/OkeyAmy/Ilera (branch ilera-atlas-pipeline) · Live demo: huggingface.co/spaces/Emmanuel-okoye/ilera-atlas

N-ATLaS fine-tuned for primary health care in Nigeria: Powered by Awarri. NAIC 2026 (National AI Innovation Challenge) · Academia & Research track · Problem Statement 3: Sectoral Fine-Tuning.

Ilera-ATLAS gives trained community health workers (CHEWs, JCHEWs, CHOs) decision support that follows Nigeria's **2024 National Standing Orders** (CHPRBN, with FMoH and NPHCDA), in **English, Hausa and Yoruba**. Every answer starts with a clear action and cites its source:

```
ACTION: TREAT | REFER | URGENT
WHAT TO DO: numbered steps
DANGER SIGNS: signs that mean urgent referral
SOURCE: document and section
```

The model is a QLoRA adapter on [NCAIR1/N-ATLaS](#) (Llama-3 8B). No other foundation model is used anywhere in the pipeline: N-ATLaS also drafts and translates the training data, which humans then review.

Pipeline

Step	Notebook	Runs on	Output (Hugging Face dataset <code>ilera-data</code>)
1. Extract the Standing Orders	01_extract	Kaggle CPU	<code>raw/chunks.jsonl</code> : 524 section-aware chunks with citations
1b. Baseline: can raw N-ATLaS find the right reference?	01b_base_reference_eval	Kaggle T4 x2 (raw float16 needs ~17 GB GPU)	<code>results/base_reference_summary_raw_fp16.csv</code> : closed-book, with-index and multiple-choice section accuracy of the unmodified model
2. Draft Q&A and case vignettes	02_draft	Kaggle T4	<code>drafts/review_en_*.csv</code> for human review
3. Human review	Google Sheets	team	<code>reviewed/review_en_final.csv</code>
4. Translate to Hausa / Yoruba	03_translate	Kaggle T4	<code>translations/review_ha_yo_*.csv</code> with quality flags
5. Freeze benchmark, build training files	03b_build	Kaggle CPU	<code>bench/ilera_bench.jsonl</code> (+SHA-256), <code>train.jsonl</code> , <code>val.jsonl</code>
6. Evaluate (baseline, run selection, final)	04_eval	Kaggle T4	<code>results/scores_*.csv</code> , paired-bootstrap comparison, blind-review sheet
7. Train (QLoRA, Unsloth)	05_train	Kaggle T4	<code>ilera-atlas-lora-run{A,B,C}</code>

Step	Notebook	Runs on	Output (Hugging Face dataset <code>ilera-data</code>)
8. Export	06_export	Kaggle T4	<code>ilera-atlas</code> (merged 16-bit), <code>ilera-atlas-GGUF</code> (Q4_K_M)
9. Serve	serve/	Modal + HF Space	OpenAI-compatible GPU API + demo with safety layer

Every notebook has a smoke-test mode, resumes after a crash, and was tested offline before use (GPU-only steps excepted). The full step-by-step guide is [docs/Ilera-ATLAS-Training-Runbook.docx](#) (its paths `ilera-atlas/notebooks/...` are `notebooks/...` in this repo); the schedule and design are in [docs/Ilera-ATLAS-Execution-Plan.docx](#).

Evaluation design

- **Ilera-Bench**: 600+ human-checked items (300 English, 150 Hausa, 150 Yoruba, parallel across languages) plus a ~60-item *unseen-section probe*, frozen with a SHA-256 hash **before** training.
- **External test**: 300 multiple-choice items from [AfriMed-QA](#) (test split, primary-care specialties, Nigerian items first).
- **Metrics**: correct action, **under-referral** (the dangerous error), over-referral, key-fact recall, correct source section, answer language (GlotLID), AfriMed-QA accuracy; paired bootstrap 95% CIs against base N-ATLaS; blind clinical review of 150 answer pairs.
- **Leak control**: no benchmark item, translation of one, probe-section text or near-duplicate question (RapidFuzz ≥ 85) enters training.

Baseline: raw N-ATLaS before fine-tuning

NCAIR1/N-ATLaS loaded **as released** (float16, no quantisation, no fine-tuning) on a Kaggle T4 x2, asked to find the Standing Orders reference for every part of the CHEW document ([01b_base_reference_eval](#), 6 Oct 2026; data in [results/](#)):

Test	Accuracy [95% CI]	Chance
Closed book: section number for a topic (105 items)	1.0% [0.0, 2.9]	0.7%
Excerpt + table of contents → section number (165 items)	9.1% [4.8, 13.3] (right title 17.0%)	0.7%
Excerpt → right section among 4 options (165 items)	75.2% [68.5, 81.8]	25%

The base model understands Standing Orders content well, but cannot produce the correct reference on its own, even with the table of contents in front of it. Ilera-ATLAS therefore (1) trains on answers that always end with the exact `SOURCE`: section and (2) is evaluated with the cited section as a metric (`source_ok`), with retrieval of the matching chunk as the planned stretch.

Results: Ilera-ATLAS vs raw N-ATLaS (frozen Ilera-Bench)

Chosen model: `ilera-atlas-lora-runB` (QLoRA $r=16$, 2 epochs, 1,058 examples; picked over the 3-epoch run on the validation set for its lower under-referral). Benchmark frozen before training (SHA-256 in [bench/](#)). Paired bootstrap, 95% CI; all differences below are significant.

Metric (496 items: 196 en + 150 ha + 150 yo)	Raw N-ATLaS	Ilera-ATLAS	Difference [95% CI]
Correct action (TREAT / REFER / URGENT)	1.8%	65.3%	+63.5 [+59.5, +67.7]
Correct action, lenient (action inferred from free text)	26.4%	65.3%	+38.9 [+33.3, +44.8]
Under-referral (dangerous error), strict	98.2%	18.5%	-79.6 [-83.3, -76.2]
Under-referral, lenient	55.8%	18.5%	-37.3 [-42.5, -32.5]
Answer format valid	8.3%	99.6%	+91.3
Key-fact recall	5.9%	23.0%	+17.1 [+14.7, +19.6]
Cites the correct Standing Orders section	0.0%	24.6%	+24.6 [+21.0, +28.4]
Answers in the question's language	80.8% (Yoruba 47.3%)	100%	+19.2

Per language, correct action: English 66.3%, Hausa 69.3%, Yoruba 60.0%. Unseen-section probe (9 items): 77.8% vs 0%. External AfriMed-QA multiple choice (300, general medicine): 38.0% vs 41.7% base, a small, non-significant drop: the model specialises in the Standing Orders. Full tables in [results/](#).

Limitations. Small training set (1,113 examples, AI-assisted review), English benchmark below the planned 300 items, AI-assisted (not yet fully clinician-verified) review of data and translations, and a blind clinical review of 150 answer pairs pending. The model can produce clinically inappropriate steps (e.g. IV medicines outside CHEW scope): it is decision support for trained workers only.

Serving (free tiers)

- **GPU API:** [serve/modal_app.py](#), vLLM on one L4, scales to zero (Modal free credits).
- **Live demo (free, always on):** huggingface.co/spaces/Emmanuel-okoye/ilera-atlas, a free static Space ([serve/static_space/](#)) that always opens the current demo. The demo itself is [notebooks/07_demo.ipynb](#): the Q4_K_M GGUF in llama.cpp on a free Kaggle CPU session (no GPU quota, ~35 s per answer) behind a Gradio share link, with the chat page [serve/space/ui.py](#) and the red-flag safety layer. It restarts itself every ~11.5 h and writes the new link to the static Space.
- **ZeroGPU Space** (when the account can host ZeroGPU, or with PRO): [serve/space/](#), the merged model on a shared GPU, same UI, plus an `/answer` API. Deploy with [serve/deploy_space.py](#).
- **API docs:** [docs/API.md](#).

Data sources

- National Standing Orders 2024 for CHEW, JCHEW and CHO: CHPRBN with FMoH and NPHCDA, via the [PHC Workforce Hub](#).
- [Aya Dataset](#) (Apache-2.0), ~10% general-language rows.
- [AfriMed-QA](#) (CC-BY-SA-4.0), evaluation only, by ID.

Licence and responsible use

Model weights derived from N-ATLaS are released under the **N-ATLaS Terms of Use** (same licence for derivatives; 1,000 active end-users per 30 days without a commercial licence). Attribution: *Awarri Technologies and the Federal Ministry of Communications, Innovation and Digital Economy*.

Ilera-ATLAS is **decision support for trained health workers**. It is not a diagnosis tool and not for patients. Medicines and doses must always be checked against the current Standing Orders.

Training configuration (notebooks/05_train.ipynb, run B)

Setting	Value
Base model	NCAIR1/N-ATLaS (Llama-3 8B), loaded in 4-bit (Unsloth)
Method	QLoRA, LoRA on q, k, v, o, gate, up, down projections
Rank / alpha / dropout	16 / 16 / 0
Learning rate, warm-up, weight decay	2e-4, 5%, 0.01
Batch	2 × 4 gradient accumulation = 8
Max sequence length	2,048 tokens
Optimiser	AdamW 8-bit; loss on responses only
Data	962 Standing Orders examples + 96 Aya general-language rows; 151 validation
Epochs, time, hardware	2 epochs, 41 min, one Kaggle T4
Outputs	LoRA adapter, merged 16-bit model, GGUF Q4_K_M (4.6 GB)

Deployment

- Free always-on demo: static Hugging Face Space (stable URL) that opens a Gradio app served by llama.cpp (Q4_K_M) on a Kaggle CPU session (notebooks/07_demo.ipynb). The session restarts itself every ~11.5 h and publishes the new link.
- Chat UI: `serve/space/ui.py` (colour-coded TREAT/REFER/URGENT cards, streaming, examples in three languages).
- Safety layer: `serve/space/redflags.csv` forces ACTION: URGENT when a danger sign is named.
- GPU options: `serve/modal_app.py` (vLLM, OpenAI-compatible) and `serve/space/app.py` (ZeroGPU Space with /answer API).

API

Two ways to call Ilera-ATLAS. Both use N-ATLaS fine-tuned on Nigeria's 2024 National Standing Orders.

Endpoint	Speed	Safety layer	Auth
GPU API (Modal, vLLM, OpenAI-compatible)	fast (first call after idle: 1–3 min cold start)	no: raw model	Authorization: Bearer <API key>
Demo Space API (Gradio)	slower on CPU fallback	yes: red-flag override + notice	Hugging Face token while the Space is private

Answer format

Every answer follows this contract (labels always in English, content in the question's language):

```
ACTION: TREAT | REFER | URGENT
WHAT TO DO: numbered steps
DANGER SIGNS: signs that mean urgent referral
SOURCE: document and section
```

Always send the system prompt from [serve/space/system_prompt.txt](#); without it the model does not use the format.
Use temperature=0.

1. GPU API (OpenAI-compatible)

curl

```
curl https://<your-modal-url>/v1/chat/completions \
-H "Authorization: Bearer $ILERA_API_KEY" -H "Content-Type: application/json" \
-d '{"model": "ilera-atlas", "temperature": 0, "max_tokens": 450, "repetition_penalty": 1.1,
  "messages": [{"role": "system", "content": "<system prompt>"},
                {"role": "user", "content": "A 2-year-old has fever and fast breathing. What should I
do?"}]}'
```

Python

```
from openai import OpenAI

client = OpenAI(base_url="https://<your-modal-url>/v1", api_key="<API key>")
system = open("serve/space/system_prompt.txt", encoding="utf-8").read()
r = client.chat.completions.create(
    model="ilera-atlas", temperature=0, max_tokens=450,
    messages=[{"role": "system", "content": system},
              {"role": "user", "content": "Mace mai ciki tana zubar da jini sosai."}],
    extra_body={"repetition_penalty": 1.1})
print(r.choices[0].message.content)
```

Parse the action with `re.search(r"ACTION:\s*(TREAT|REFER|URGENT)", text)`.

2. Demo Space API (with the safety layer)

```
from gradio_client import Client

c = Client("<you>/ilera-atlas-demo", hf_token="hf_...") # hf_token only needed while the Space is private
print(c.predict("A pregnant woman has a severe headache and blurred vision.", api_name="/answer"))
```

Limits and responsible use

- Decision support for **trained health workers**; not for patients; not a diagnosis.
- N-ATLaS licence: at most **1,000 active end-users per 30 days** without a commercial licence from Awarri / FMCIDE. Attribution: *Awarri Technologies and the Federal Ministry of Communications, Innovation and Digital Economy*. Powered by Awarri.

- Do not send personal or identifying patient data.